

Lycanthrope Template Implementation Analysis

Date: 2026-05-23
Project: /home/ubuntu/dnd35prototype
Status:  Fully Implemented — Quality improvements possible

1. Current Implementation Status

Files & Line Counts

File	Lines	Purpose
Assets/Scripts/Character/LycanthropeTemplate.cs	1,124	Template logic + Factory (8 variants) + Enum + Helpers
Assets/Scripts/Character/NPCDatabase_Lycanthropes.cs	95	Registration + 7 encounter presets
Assets/Scripts/Effects/Disease.cs	+1 line	Lycanthropy added to DiseaseType enum
Assets/Scripts/Character/NPCDatabase.cs	+2 lines	RegisterCreatures_Lycanthropes() in Init, GetLycanthropeEncounterPresets() in presets
Total	~1,222	

Brace Balance:  **Both files balanced (0 offset)**

Classes/Types Defined

Type	Name	Purpose
static class	LycanthropeTemplate	Core template logic (Apply() + helpers)
struct	AnimalFormModifiers	Data container for animal form stats
enum	LycanthropeAnimalType	7 animal types (Wolf, Dire-Wolf, Rat, Boar, DireBoar, Tiger, BrownBear)
static class	LycanthropeFactory	8 pre-defined variants + CreateFromRegistered()

Registered Variants (8 total)

ID	Name	Base Creature	Animal	CR	Natural?
werewolf	Werewolf (Hybrid)	Human Warrior 1	Wolf	3	
were-wolf_lord	Werewolf Lord (Hybrid)	Human Fighter 10	Dire Wolf	14	
wererat	Wererat (Hybrid)	Human Rogue 1	Dire Rat	2	
wereboar	Wereboar (Hybrid)	Human Barbarian 1	Boar	4	
weretiger	Weretiger (Hybrid)	Human Noble 4	Tiger	8	
werebear	Werebear (Hybrid)	Human Commoner 1	Brown Bear	5	
dire_wereboar	Dire Wereboar (Hybrid)	Human Barbarian 1	Dire Boar	7	
were-wolf_afflicted	Afflicted Werewolf (Hybrid)	Human Commoner 1	Wolf	2	

Encounter Presets (7 total)

ID	Name	EL	Description
werewolf_pack	 Werewolf Pack	5	3 werewolves
wererat_ambush	 Wererat Ambush	5	4 wererats
wereboar_rampage	 Wereboar Rampage	6	2 wereboars
weretiger_hunt	 Weretiger Hunt	8	Solo weretiger
lycan-thrope_menagerie	 Lycanthrope Menagerie	8	Mixed showcase
were-wolf_lord_encounter	 Werewolf Lord	15	Boss + wolf escorts
cursed_villagers	 Cursed Villagers	4	3 afflicted werewolves

2. D&D 3.5e Rules Accuracy

Stat Verification Results: ALL 6 TESTED VARIANTS PASS

Automated verification confirmed all hybrid form stat calculations match MM expectations:

Variant	STR	DEX	CON	WIS	Total HD	BAB	Nat Armor	DR
Werewolf	15 	15 	16 	13 	3 	2 	4 	10/silver 
Wererat	10 	21 	14 	14 	2 	0 	3 	10/silver 
Wereboar	20 	10 	20 	12 	4 	3 	8 	10/silver 
Weretiger	26 	20 	18 	14 	10 	8 	5 	10/silver 
Werebear	30 	12 	22 	14 	7 	4 	7 	10/silver 
Afflicted WW	14 	15 	15 	12 	3 	1 	4 	5/silver 

Rules Implemented

Rule	Status	Details
Type: keep base + Shapechanger subtype	✓	Base type preserved, "Shapechanger" tag added
HD = creature HD + animal HD	✓	Combined correctly for all variants
Combined BAB	✓	creature BAB + animal BAB
Physical ability modifiers from animal	✓	STR/DEX/CON modifiers per MM p.175 table
WIS +2 all lycanthropes	✓	Applied universally
Natural armor = max(creature, animal) + 2	✓	Correct formula
DR 10/silver (natural)	✓	
DR 5/silver (afflicted)	✓	
Hybrid form: 2 claws + bite	✓	Correct natural attacks with size-appropriate damage
Bite = secondary (half STR)	✓	BonusDamage-Source.StrengthHalf
Iron Will bonus feat	✓	Added if not already present
Scent in all forms	✓	HasScent = true
Curse of Lycanthropy (natural only)	✓	HasDiseaseOnHit = true, DiseaseType.Lycanthropy on bite
Afflicted cannot spread curse	✓	No disease on hit when isNatural = false
Trip on wolf/dire wolf bite	✓	HasTripAttack = true with bonus
Pounce for weretiger hybrid	✓	HasPounce = true (tiger only)
Improved grab (animal form only)	✓	Correctly NOT carried to hybrid
Rake (animal form only)	✓	Correctly NOT carried to hybrid

Rule	Status	Details
Hybrid size = larger of creature/animal	✓	<code>Mathf.Max()</code> on <code>SizeCategory</code>
CR calculation by animal HD	✓	MM p.175 table (1-2 HD: +2, 3-5: +3, 6-10: +4)
Low-light vision	⚠	Listed in <code>SpecialAbilities</code> text but not mechanically tracked (no system exists)
Lycanthropic empathy	⚠	Listed in <code>SpecialAbilities</code> text but not mechanically tracked

Rules NOT Implemented (Design Choices)

Rule	Status	Reason
Three-form system (human/animal/hybrid)	✗	Only hybrid form registered — correct for combat encounters
Alternate Form as standard action	✗	No form-shifting combat mechanic — noted in abilities text
Control shape checks (afflicted)	✗	No transformation control system — would need new mechanic
Lycanthropic empathy checks	✗	No social skill system in combat prototype
Alignment restrictions per type	⚠	Not enforced mechanically (werebear=LG, werewolf=CE, etc.)
Type change to “Magical Beast” in animal form	N/A	Only hybrid form exists
Vulnerability to removal of lycanthropy	N/A	No curse removal system

3. Quality Score

Metric Breakdown (1-10)

Metric	Score	Notes
D&D Rules Accuracy	9	All verifiable hybrid stats correct. Missing only non-combat flavor rules.
Code Organization	8	Clean separation: Template → Factory → Registration. Good XML docs.
Extensibility	7	Easy to add new animal types via <code>LycanthropeAnimalType</code> enum + <code>GetAnimalModifiers()</code> .
Code Reuse	6	No <code>StripSpecialEffects</code> duplication (handles attacks differently). Some base creature def duplication with Skeleton/Zombie.
Documentation	9	MM page references, inline stat verification comments, comprehensive XML docs.
Integration Quality	7	Disease system integrated. <code>NPCDatabase</code> registration correct. No runtime application support.
Error Handling	7	Null checks, fallback defaults in enum switch, <code>Debug.LogError</code> for unknowns.
Testing Infrastructure	8	7 encounter presets covering all CR ranges. Stat verification comments inline.
Composability	3	Cannot compose with other templates. No <code>ICreatureTemplate</code> integration.
Runtime Application	2	Build-time only. Cannot apply to arbitrary creatures at runtime.
Multi-Form Support	2	Only hybrid form. No form-switching mechanic.

Metric	Score	Notes
Overall	6.2	

Comparison to Other Templates

Metric	Celestial/Fiendish	Skeleton/Zombie	Lycanthrope
D&D Rules Accuracy	9	9	9
Modularity	9	6	7
Extensibility	9	7	7
Code Reuse	8	4	6
Composability	8	2	3
Runtime Application	9	2	2
Multi-Form Support	N/A	N/A	2
Overall	8.3	6.3	6.2

4. Specific Issues

4.1 No `ICreatureTemplate` Integration (Medium Priority)

The Lycanthrope template doesn't participate in the `CreatureTemplateRegistry`. This means:

- Cannot be applied at runtime (e.g., if a PC gets cursed and transforms)
- Cannot be composed with other templates (e.g., "Fiendish Werewolf")
- `AppliedTemplateIds` is set to `{ "lycanthrope", "[prefix]" }` but the registry doesn't know about it

4.2 No `StripSpecialEffects` Duplication (✅ Not an Issue)

Unlike Skeleton/Zombie, the Lycanthrope template handles natural attacks by **replacing them entirely** (creates fresh claw + bite attacks for hybrid form) rather than stripping effects from existing ones. This is actually the **correct approach** for lycanthropes per MM — the hybrid form gets its own natural attacks, not modified versions of the base creature's.

4.3 Base Creature Definition Duplication (Low-Medium Priority)

The Lycanthrope factory defines base creatures inline that overlap with other templates:

- `base_human_warrior` — also in `SkeletonFactory` and `ZombieFactory` (similar but not identical stats)
- `base_human_commoner` — also in `ZombieFactory` (similar)
- `base_human_barbarian` — unique to Lycanthrope (2 variants: wereboar + dire wereboar)

Note: The Lycanthrope base creatures are **not identical** to the Skeleton/Zombie ones — they have different ability scores, equipment, feats, and BaseHitDieHP. This is intentional — the MM lists different

base creatures for each template example. So this is less “duplication” and more “similar-but-distinct base creatures.”

4.4 Only Hybrid Form Registered (Design Choice, Not Bug)

The template only builds **hybrid form** stat blocks. This is noted in the code comments:

“Our implementation registers the HYBRID form stat block since that’s the most common combat encounter form.”

For a combat prototype, this is correct. You almost always encounter lycanthropes in hybrid form during combat. The human form is for RP/social encounters, and the animal form is a secondary combat option. However, if the game adds:

- PC lycanthropy (curse system)
- NPC form-shifting during combat
- Voluntary/involuntary transformation triggers

...then multi-form support would be needed.

4.5 Disease System Integration (Partial)

The Lycanthropy curse is implemented via the disease system:

- `DiseaseType.Lycanthropy` exists in the enum
- Bite attacks on natural lycanthropes set `HasDiseaseOnHit = true` + `DiseaseOnHitType = DiseaseType.Lycanthropy`
- The combat system (`CharacterController.cs` line 4501) properly calls `target.ExposeToDisease()` on hit

What’s missing:

- No disease resolution for Lycanthropy specifically (no Fort DC 15 save implementation for this specific disease)
- No mechanical effects after contracting Lycanthropy (no involuntary transformation, no alignment shift)
- No cure/removal system (Belladonna, Remove Disease within 3 days, etc.)

The disease hookup is a **placeholder** — it fires the exposure event but the downstream effects aren’t lycanthropy-specific yet.

4.6 AI Archetype Limitations (Minor)

The template uses:

- `NPCAIPProfileArchetype.Berserk` for most lycanthropes
- `NPCAIPProfileArchetype.Humanoid` for wererat

The enum doesn’t have a `Lycanthrope` or `ShapeChanger` archetype. The current choices are reasonable (berserk = charge-and-attack, humanoid = tactical flanking) but wouldn’t handle form-switching AI.

4.7 No Alignment Enforcement (Minor)

Per MM, each lycanthrope type has a specific alignment tendency:

- Werewolf: CE (Chaotic Evil)
- Wererat: LE (Lawful Evil)
- Wereboar: N (True Neutral)
- Weretiger: N (True Neutral)
- Werebear: LG (Lawful Good)

The template doesn't set `CharacterAlignment` — it inherits from the base creature. This matters for Smite Evil/Good targeting and Protection from Evil/Good effects.

5. What the Lycanthrope Template Does Better Than Skeleton/Zombie

1. **No `StripSpecialEffects` duplication** — Creates fresh attacks instead of stripping existing ones
2. **More complex template logic** — Merges two creature stat blocks (creature + animal), not just converting one
3. **Data-driven animal types** — `AnimalFormModifiers` struct + `GetAnimalModifiers()` switch makes adding new animal types clean
4. **Natural vs Afflicted handling** — Single `isNatural` parameter cleanly controls DR amount and curse spreading
5. **Proper disease system integration** — Uses existing `DiseaseOnHitType` on bite attacks
6. **More varied factory** — 8 variants spanning CR 2-14 with different base classes and equipment

6. What Celestial/Fiendish Does Better

1. **`ICreatureTemplate` interface** — Lycanthrope uses static class pattern
2. **Registry participation** — Lycanthrope can't be resolved by string ID at runtime
3. **Runtime application** — Can apply Celestial/Fiendish to any creature dynamically
4. **Shared base class** — `OutsiderTemplateBase` eliminates duplication; Lycanthrope has no base class
5. **Combat action system** — `TemplateSmiteSystem.cs` (247 lines) is a full combat action with UI, AI, targeting; Lycanthrope has no equivalent dedicated combat system
6. **Template composition** — `AppliedTemplateIds` list supports stacking; Lycanthrope is standalone

7. Refactoring Plan: Bringing Lycanthrope to Celestial/Fiendish Quality

Phase A: `ICreatureTemplate` Wrapper (Effort: ~1 hour)

Create a thin wrapper that implements `ICreatureTemplate` :

```
// Assets/Scripts/Character/Templates/LycanthropeCreatureTemplate.cs
public sealed class LycanthropeCreatureTemplate : ICreatureTemplate
{
    public string TemplateId => "lycanthrope";

    // Default to wolf hybrid for runtime application
    private LycanthropeAnimalType _animalType;
    private bool _isNatural;

    public LycanthropeCreatureTemplate(
        LycanthropeAnimalType animalType = LycanthropeAnimalType.Wolf,
        bool isNatural = true)
    {
        _animalType = animalType;
        _isNatural = isNatural;
    }

    public void ApplyToDefinition(NPCDefinition definition)
    {
        // Apply in-place since ApplyTemplatesClone already cloned
        NPCDefinition result = LycanthropeTemplate.Apply(definition, _animalType, _isNatural);
        if (result != null)
            definition.CopyFieldsFrom(result);
    }
}
```

Challenge: The Lycanthrope template requires an `animalType` parameter that the generic `ICreatureTemplate.ApplyToDefinition(NPCDefinition)` interface doesn't support. Options:

1. **Default animal type** — Use Wolf as default, override via `AppliedTemplateIds` naming convention (e.g., `"lycanthrope_wolf"`, `"lycanthrope_tiger"`)
2. **Extended metadata** — Add an optional `TemplateMetadata` dictionary to `NPCDefinition` that templates can read
3. **Per-animal-type templates** — Register each as a separate template: `"werewolf"`, `"wererat"`, `"wereboar"`, etc.

Recommendation: Option 3 is cleanest — register each lycanthrope type as its own template ID:

```
{ "werewolf", new LycanthropeCreatureTemplate(LycanthropeAnimalType.Wolf, true) },
{ "wererat", new LycanthropeCreatureTemplate(LycanthropeAnimalType.Rat, true) },
{ "wereboar", new LycanthropeCreatureTemplate(LycanthropeAnimalType.Boar, true) },
{ "weretiger", new LycanthropeCreatureTemplate(LycanthropeAnimalType.Tiger, true) },
{ "werebear", new LycanthropeCreatureTemplate(LycanthropeAnimalType.BrownBear, true) }
,
```

Phase B: Multi-Form System (Effort: ~4-6 hours)

This is the **biggest gap** and the hardest to close. Full multi-form support would require:

B.1: Form Data Structure

```
public class LycanthropeFormData
{
    public NPCDefinition HumanoidForm; // base creature stats
    public NPCDefinition AnimalForm;   // full animal stats
    public NPCDefinition HybridForm;   // merged stats (what we have now)
    public LycanthropeAnimalType AnimalType;
    public bool IsNatural;
}
```

B.2: Form-Switching Combat Action

Similar to `TemplateSmiteSystem.cs`, create `LycanthropeFormShiftSystem.cs`:

```
public partial class GameManager
{
    public bool CanShiftForm(CharacterController actor, out string reason) { ... }
    public void OnShiftFormButtonPressed() { ... }
    public void ExecuteFormShift(CharacterController actor, LycanthropeForm target-
Form) { ... }
}
```

- Shifting is a standard action (or move-equivalent for natural lycanthropes at higher levels)
- Replace the NPC's stat block with the appropriate form's stats
- Update visual appearance (sprite color)
- Trigger Fortitude save for afflicted lycanthropes (Control Shape DC 25)

B.3: Involuntary Transformation System

- Full moon trigger (if time tracking exists)
- Damage threshold trigger (when below 50% HP in human form)
- Control Shape check (WIS-based, DC 25 to resist)
- Berserk mode when involuntarily transformed

B.4: AI Form Selection

Add form-switching logic to AI:

- If in human form and combat starts → shift to hybrid
- If low HP and in animal form → consider fleeing
- Afflicted: may involuntarily shift if Control Shape fails

Recommendation: This is a **significant new system**. Only implement when:

- PC lycanthropy curse is added
- NPC AI needs to shift forms during combat
- Multi-phase boss encounters are desired

Phase C: Curse System Deep Integration (Effort: ~2-3 hours)

C.1: Lycanthropy-Specific Disease Resolution

```
// In Disease resolution system:
case DiseaseType.Lycanthropy:
    // DC 15 Fort save
    // If failed: creature contracts lycanthropy
    // Incubation: next full moon
    // Effect: involuntary transformation
    break;
```

C.2: Belladonna Cure

```
// Within 1 hour of contracting: DC 20 Heal check
// Within 3 days: Remove Disease (CL 12+)
// After 3 days: Remove Curse (CL 12+) or Heal spell
```

C.3: PC Transformation

If a PC contracts lycanthropy:

- After incubation, involuntary shift during full moon
- Must make Control Shape checks
- Can eventually learn to control transformations
- Becomes natural lycanthrope after 3 full moons

Recommendation: Only implement when the game has a disease/curse resolution system with meaningful consequences.

Phase D: Alignment Integration (Effort: ~30 min)

Add alignment to factory-created lycanthropes:

```
lycan.CharacterAlignment = CharacterAlignment.ChaoticEvil; // werewolf
lycan.CharacterAlignment = CharacterAlignment.LawfulEvil;  // wererat
lycan.CharacterAlignment = CharacterAlignment.TrueNeutral; // wereboar
lycan.CharacterAlignment = CharacterAlignment.TrueNeutral; // weretiger
lycan.CharacterAlignment = CharacterAlignment.LawfulGood;  // werebear
```

This enables Smite Evil/Good, Protection from Evil, etc. to work correctly against lycanthropes.

8. Effort Estimates

Phase	Effort	Priority	Unlocks
A: ICreatureTemplate wrapper	~1 hour	Medium	Runtime application, registry integration
B: Multi-form system	~4-6 hours	Low	Form-shifting combat, NPC AI forms
C: Curse deep integration	~2-3 hours	Low	PC lycanthropy, disease resolution
D: Alignment integration	~30 min	Medium	Smite/Protection targeting
Total to match Celestial/Fiendish	~8-10 hours		

Quick Wins (< 1 hour total):

1. ☒ Phase D: Add alignment to factory creatures (~30 min)
2. ☒ Phase A: Create ICreatureTemplate wrapper class (~1 hour)

Do When Needed:

1. Phase C: Curse resolution — when disease system is expanded
2. Phase B: Multi-form — when PC lycanthropy or NPC form-shifting is needed

9. Unique Challenges for Lycanthrope Template

9.1 Three-Form System

Unlike other templates (which are one-way transformations), lycanthropes have **three forms** that they actively switch between during play:

Form	Stats	Natural Attacks	Equipment	AI
Humanoid	Base creature only	None (uses weapons)	Full equipment	Normal
Hybrid	Merged (what we have)	2 claws + bite + weapons	Can use weapons	Aggressive
Animal	Animal stats + template bonuses	Animal attacks only	Cannot use weapons	Animal-like

This is fundamentally different from Skeleton (one-way conversion), Zombie (one-way conversion), or Celestial/Fiendish (stat overlay).

9.2 Template Requires TWO Inputs

The `ICreatureTemplate.ApplyToDefinition(NPCDefinition)` interface takes a single creature definition. But Lycanthrope needs **two inputs**: a base creature AND an animal type. This is a design mismatch that requires either:

- Per-animal template IDs (recommended)
- Extended metadata on the NPCDefinition
- A more complex interface

9.3 Disease Transmission Creates New Lycanthropes

Unlike other templates, the Lycanthrope template can **create copies of itself** through the Curse of Lycanthropy. A werewolf bites a PC → PC becomes a werewolf → PC can now bite others. This recursive property doesn't exist in any other template.

9.4 Alignment Variation

Each lycanthrope type has its own alignment (CE, LE, N, LG), unlike Skeleton (always NE), Zombie (always NE), or Celestial/Fiendish (always Good/Evil). This means alignment-based systems need to check the specific lycanthrope type.

9.5 Natural vs Afflicted is a Spectrum

The natural/afflicted distinction isn't just DR 10 vs 5 — it affects:

- Curse spreading ability
- Transformation control
- Mental stats in animal form
- Ability to have children who are natural lycanthropes
- Whether they can be cured by Remove Disease

This makes the template more of a **character progression system** than a simple stat overlay.

10. Conclusion

Current State: Good Foundation, Room for Growth

The Lycanthrope template is a **solid combat encounter system** that accurately implements D&D 3.5e hybrid form stats for 8 variants spanning CR 2-14. It integrates with the disease system for curse transmission and follows the established Skeleton/Zombie factory pattern.

To Match Celestial/Fiendish Quality:

The gap is **not just about code architecture** — it's about the fundamental complexity difference:

- Celestial/Fiendish is a **stat overlay** (one creature → enhanced creature)
- Lycanthrope is a **creature merger** (two creatures → hybrid creature)

Matching the architectural quality (ICreatureTemplate, registry, runtime application) is ~1-2 hours. But matching the **gameplay depth** (multi-form, curse progression, form-shifting AI) is ~6-8 hours of new system development.

Recommended Path:

1. **Now:** Add alignment to factory variants (30 min) ← Quick win
2. **Soon:** Create `ICreatureTemplate` wrapper with per-animal-type registration (1 hour) ← Enables runtime use
3. **When needed:** Multi-form system (4-6 hours) ← When game needs form-shifting
4. **When needed:** Deep curse integration (2-3 hours) ← When disease system matures

The Lycanthrope template is **6.2/10 quality** vs Celestial/Fiendish at **8.3/10**. The Quick Wins can bring it to **~7.0/10**; full multi-form support would bring it to **~8.5/10** (potentially higher than Celestial/Fiendish due to the added gameplay depth).